HPE juniper networking

Data sheet

JUNIPER NETWORKS EX2300-C SWITCH

HPE

Product overview

The Juniper Networks EX2300-C Switch offers an economical, entry-level solution in a compact, fanless form factor for access layer deployments in branches, retail, and workgroup environments. The EX2300-C supports HPE Juniper Networking Virtual Chassis technology, allowing up to four interconnected switches to be managed as a single logical device, delivering a scalable, pay-as-you-grow solution for expanding networks. The EX2300-C is onboarded, provisioned, and managed in the Mist platform's cloud, with Wired Assurance delivering better experiences for connected devices through AI-powered automation and service levels.

Product description

The Juniper Networks EX2300-C Switch delivers a compact, silent, and powerefficient platform for low-density branch deployments and commercial access or enterprise workgroup environments outside the wiring closet.

Featuring 12 10/100/1000BASE-T access ports and two 10GbE uplink ports with and without Power over Ethernet (PoE/PoE+) in a fanless design, the EX2300-C switches deliver a powerful solution for supporting services such as unified communications, IP telephony, closed circuit television (CCTV), and other applications in office, classroom, hospitality, and other space and wiring-constrained environments. The key features of the EX2300-C switch include:

— Multiple hardware configurations:

- 12 10/100/1000BASE-T access ports and two SFP+ 10GbE uplink ports
- 12 10/100/1000BASE-T PoE/PoE+ access ports and two SFP+ 10GbE uplink ports

— IEEE 802.3at (PoE+) compliance, allowing a variety of devices to draw power from the switch through the access ports

— Cloud-managed and driven by Mist AI with Juniper Wired Assurance

— Virtual Chassis support, allowing up to four switches to be interconnected and managed as a single logical device (optional license required)

— Energy Efficient Ethernet (EEE) support for GbE access ports

— Compact design featuring fanless and noiseless operation

Architecture and key components

The fixed-configuration EX2300-C line of Ethernet switches delivers complete Layer 2 and basic Layer 3 switching capabilities to satisfy the low-density branch and low-density wiring closet connectivity requirements of today's high performance business needs. The EX2300-C models extend that reach into workgroup environments that reside outside the wiring closet, including branch offices, retail access applications, education, hospitality, and# other locations where the switch is deployed in open areas inside a building.

Two versions are available. The EX2300-C-12T offers 12 front panel 10/100/1000BASE-T access ports and two SFP+ 10GbE uplink ports. The EX2300-C-12P offers 12 front panel 10/100/1000BASE-T IEEE 802.3af/IEEE 802.3at (PoE/PoE+) access ports, which can be used for powering networked devices such as telephones, video cameras, multiple radio IEEE 802.11ac wireless LAN (WLAN) access points, and videophones in converged network environments. In addition, the EX2300-C-12P has two SFP+ 10GbE uplink ports.

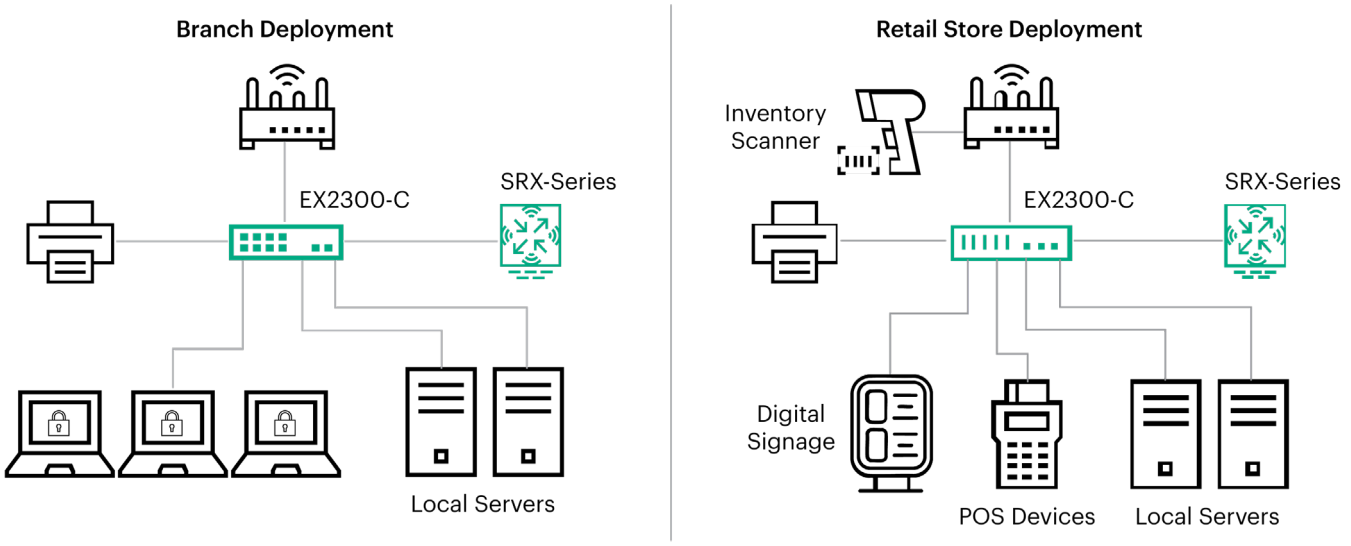

Figure 1. EX2300-C branch and retail store deployments.

The uplink ports can be used to connect to higher-layer devices such as aggregation switches or routers. The uplink ports can also be configured as Virtual Chassis interfaces and connected via standard 10GbE interfaces. A fixed internal power supply ensures operational simplicity.

Cloud management with Juniper Wired Assurance

Juniper Wired Assurance, a cloud-based service driven by Mist AI to claim, configure, manage, and troubleshoot the EX2300-C, delivers AI-powered automation and service levels to ensure a better experience for connected devices. Wired Assurance leverages rich Junos switch telemetry data to simplify operations, reduce mean time to repair, and improve visibility. Wired Assurance offers the following features:

— **Day 0 operations—**Onboard switches seamlessly by claiming a greenfield switch or adopting a brownfield switch with a single activation code for true plug-and-play simplicity.

— **Day 1 operations—**Implement a template-based configuration model for bulk rollouts of traditional and campus fabric deployments, while retaining the flexibility and control required to apply custom site- or switch-specific attributes. Automate provisioning of ports via Dynamic Port Profiles.

— **Day 2 operations—**Leverage the AI in Juniper Wired Assurance to meet service-level expectations such as throughput, successful connects, and switch health with key pre- and post-connection metrics (see Figure 2). Add the self-driving capabilities in Marvis Actions to detect loops, add missing VLANs, fix misconfigured ports, identify bad cables, isolate flapping ports, and discover persistently failing clients (see Figure 3). And perform software upgrades easily through Mist platform's cloud.

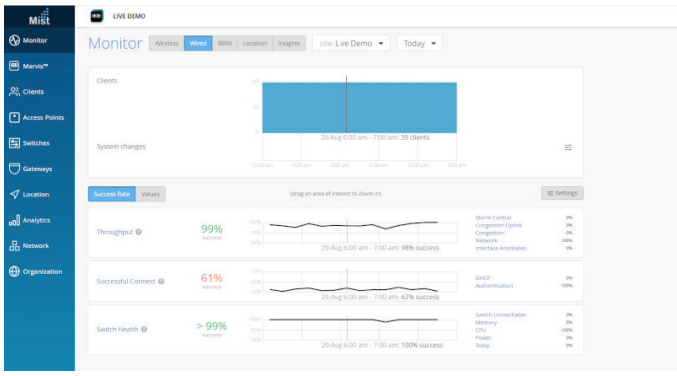

Figure 2. Juniper Wired Assurance service-level expectations

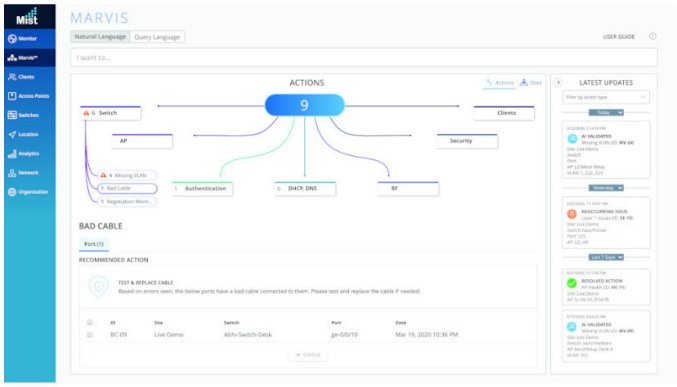

Figure 3. Marvis Actions for wired switches

The addition of Marvis, a complementary Virtual Network Assistant driven by Mist AI, lets you start building a self-driving network that simplifies network operations and streamlines troubleshooting via automatic fixes for EX Series switches or recommended actions for external systems.

For more information see Juniper Wired Assurance.

Virtual Chassis technology

The EX2300-C supports Juniper's unique Virtual Chassis technology, enabling up to four interconnected EX2300-C switches to be managed as a single logical device and delivering a scalable, pay-as-you-grow solution for expanding network environments. The EX2300-C can also be connected to an existing Virtual Chassis configuration composed of EX2300 switches.

While EX2300-C switches can be interconnected over either of the two front panel 10GbE uplink ports, these ports can also be configured as 1GbE uplinks to aggregation devices by disabling the Virtual Chassis technology.

When deployed in a Virtual Chassis configuration, the EX2300-C switches elect a primary and a backup switch based on a set of preconfigured policies or criteria. The primary switch automatically creates and updates the switching and optional routing tables on all other Virtual Chassis switch members. Switches can be added to or removed from the Virtual Chassis configuration without service disruption.

Simplified operations

Virtual Chassis technology simplifies network management for smaller deployments. Up to four interconnected EX2300-C switches can be managed as a single device utilizing a single Junos OS image and a single configuration file, reducing the overall number of units to monitor and manage. When the Junos OS is upgraded on the primary switch in an EX2300-C Virtual Chassis configuration, the software is automatically upgraded on all other member switches at the same time.

Dedicated front panel RJ-45 and USB console ports offer flexible out-of-band management options, while a front panel USB port can be used to easily upload the HPE Juniper Networking Junos operating system and configuration files. All front panel access and uplink ports have dedicated link status and link activity LEDs. In addition, a series of front panel LEDs offer system status information with a button that allows users to toggle through four different modes for reporting speed (SPD), duplex mode (DX), administrative status (EN), and PoE status (PoE). A fixed internal power supply ensures operational simplicity.

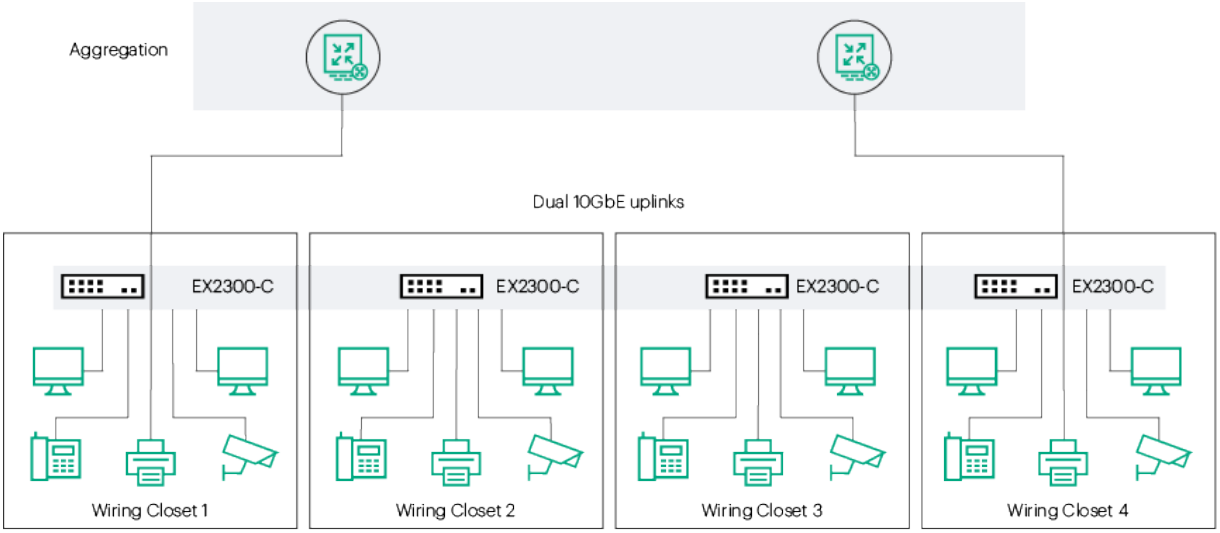

Figure 4. When deployed in a Virtual Chassis configuration, up to four EX2300-C Switches can operate as a single, logical device.

EX2300-C Virtual Chassis configurations operate as highly resilient unified systems, providing simplified management using a single IP address, single telnet/SSH session, single command-line interface (CLI), automatic version checking, and automatic configuration. The EX2300-C switches are also capable of local switching, so packets coming into a port destined for another port on the same switch do not have to traverse the Virtual Chassis, increasing forwarding capacities.

EX2300-C Virtual Chassis configurations implement the same slot/ module/port numbering schema as other HPE Juniper Networking chassis-based products, providing true chassis-like operations. By using a consistent operating system and a single configuration file, all switches in a Virtual Chassis configuration are treated as a single device, simplifying overall system maintenance and management.

Easy provisioning

Auto-configuration and Auto-Image Install features allow the switches to be configured and imaged over the network using the Dynamic Host Configuration Protocol (DHCP) message exchange process. These features eliminate the need to stage devices prior to deployment, dramatically reducing operational expenses. Autoconfiguration and Auto Image Install allow new branches and retail stores to rapidly deploy their networks, and they also enable software upgrades and security fixes to be performed at the touch of a button. The ZTP feature allows a DHCP server to push configuration details and software images to multiple switches at boot-up time.

Campus fabric deployments

Juniper campus fabrics support these validated architectures with the EX2300-C switch playing the role of access switch in a Virtual Chassis:

— **EVPN multihoming (collapsed core or distribution):** A collapsed core architecture combines the core and distribution layers into a single switch, turning the traditional three-tier hierarchal network into a two-tier network. This eliminates the need for STP across the campus network by providing multihoming capabilities from the access to the core layer. EVPN multihoming can be deployed and managed using the Mist platform's cloud.

— **Core-distribution:** A pair of interconnected EX Series core or distribution switches provide L2 EVPN and L3 VXLAN gateway support. The EVPN-VXLAN network between the distribution and core layers offers two modes: centrally or edge routed bridging overlay.

In all these EVPN-VXLAN deployment modes, EX2300-C switches can be used at the access layer.

Features and benefits

The EX2300-C model switches include a number of features that make them ideal for low-density, open space deployments.

Managing AI-driven campus fabric with the Mist platform's cloud

Juniper Wired Assurance brings cloud management and Mist AI to campus fabric. It sets a new standard moving away from traditional network management towards AI-driven operations, while delivering better experiences to connected devices. The Mist platform's cloud streamlines deployment and management of campus fabric architectures by allowing:

— Automated deployment and zero touch deployment

— Anomaly detection

— Root cause analysis

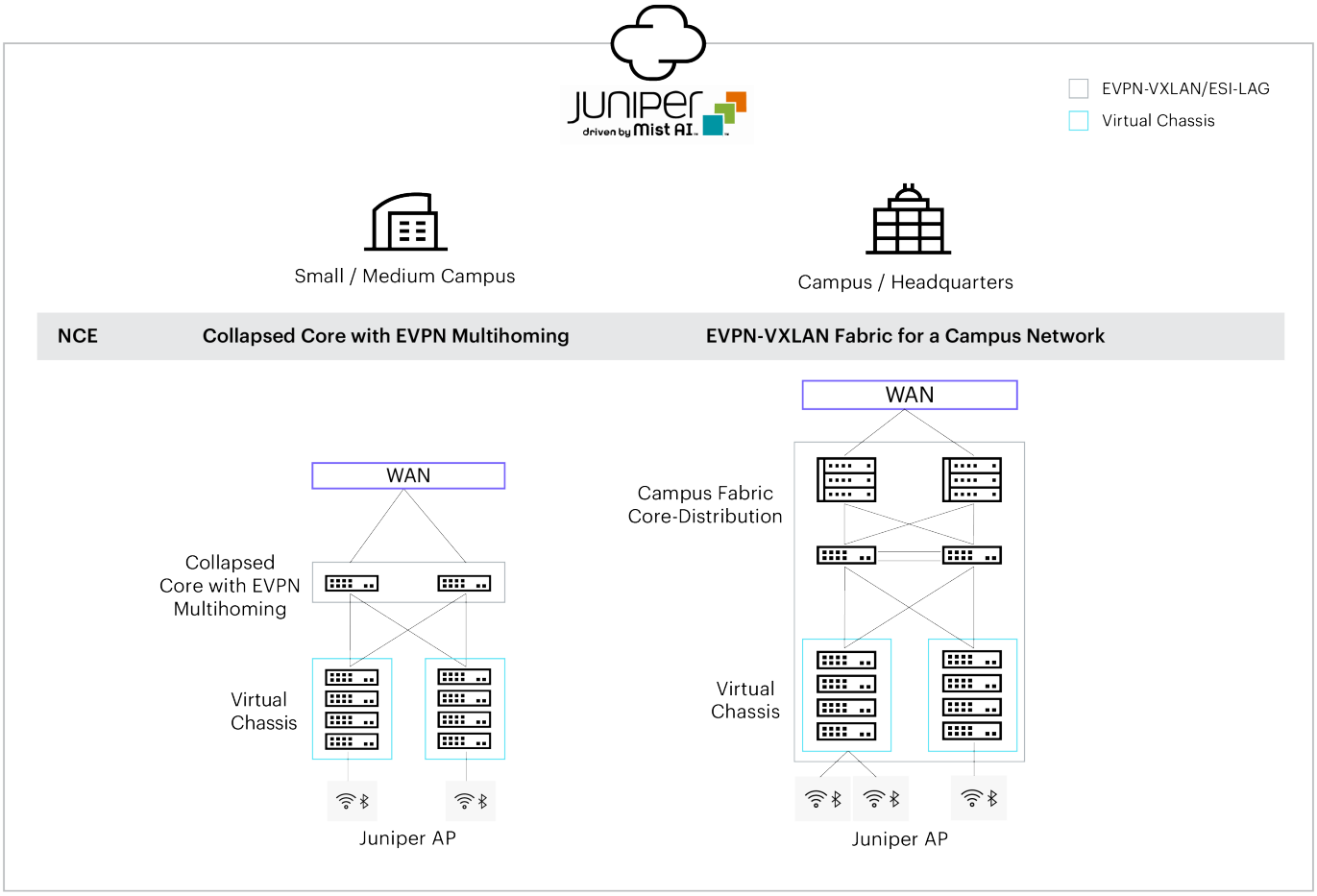

Figure 5. Campus fabrics showing Virtual Chassis and EVPN-VXLAN-based architectures

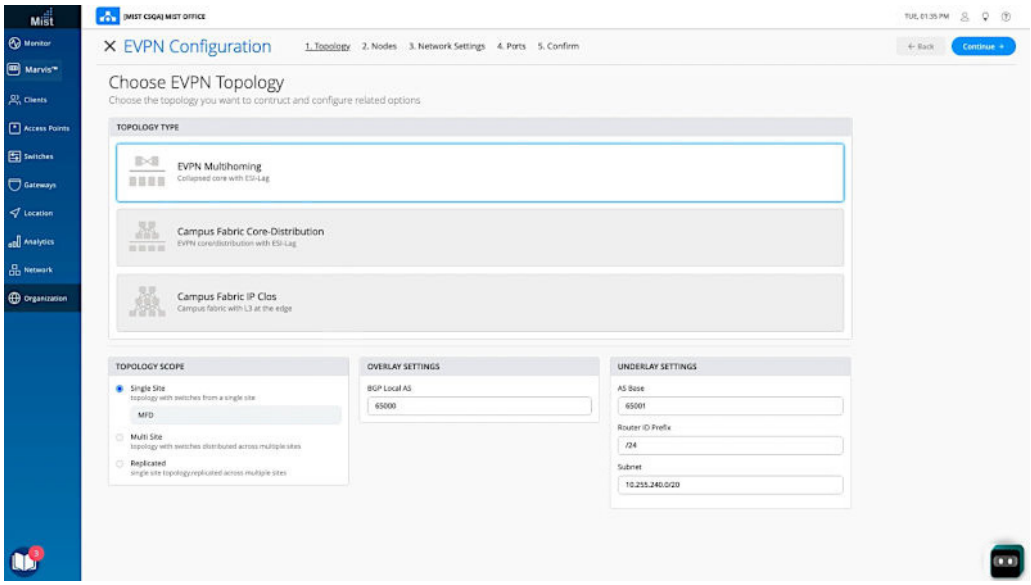

Figure 6. EVPN multihoming configuration via the Mist platform's cloud

Compact form factor

At 10.98 inches wide and 9.4 inches deep, EX2300-C switches can be easily and discreetly installed on desktops, on or under shelves, or on walls. An optional magnetic mounting pad is available for securing the EX2300-C switches to metal surfaces. Rack-mount kits are also available for installing the switches in standard 19-inch wiring racks. Wall mounting can be done using the flexible mounting slots on the bottom of the chassis to attach to the screws on the wall.

Silent operation

The EX2300-C switches are fanless, resulting in a silent operation suitable for deployments in workgroup areas. The fanless design also reduces power consumption and improves mean time between failures (MTBF) by eliminating moving parts.

Low power

The fanless design reduces the power consumed by the EX2300-C switches. Energy Efficient Ethernet (EEE) ports ensure low power consumption when the network link is idle.

Access security

Working as an enforcement point within the Access Policy Infrastructure, the EX2300-C provides standards-based 802.1X port-level access control as well as L2-L4 policy enforcement based on user identity, location, device, or a combination of these. If access is granted, the switch assigns the user to a specific VLAN based on authorization levels.

The EX2300-C also provides a full complement of port security features, including DHCP snooping, dynamic ARP inspection (DAI), and media access control (MAC) limiting to defend against internal and external spoofing, and man-in-the-middle and denial-of-service (DoS) attacks.

Physical security

Security slots on either side of the switch accept locking devices that physically secure switches, preventing them from being easily removed in open space or unsecured environments.

PoE/PoE+ power

The EX2300-C-12P switch delivers power for supporting networked devices such as telephones, video cameras, IEEE 802.11n WLAN access points, and videophones. It supports the IEEE 802.3af PoE standard, as well as IEEE 802.3at PoE+ with a budget of 124 watts. At maximum power, the 12-port EX2300 C-12P can provide the full 15.4 watts of Class 3 PoE to a maximum of eight ports simultaneously, and the full 30 watts of PoE+ to a maximum of four ports simultaneously. Attached devices draw the necessary power until the PoE power budget is exhausted.

Junos operating system

The EX2300-C switches run the same Junos OS that is used by other Juniper Networks Ethernet Switches, QFX Series Switches, HPE Juniper Networking Routers, Juniper Networks SRX Series Firewalls, and the Juniper Networks NFX Series Network Services Platform. By using a common operating system, HPE delivers a consistent implementation and operation of control plane features across all products.

Flex licensing

Flex licensing offers a common, simple, and flexible licensing model for EX Series access switches, enabling customers to purchase features based on their network and business needs.

Flex licensing is offered in Standard, Advanced, and Premium tiers. Standard tier features are available with the Junos OS image that ships with EX Series switches. Additional features can be unlocked with the purchase of a Flex Advanced or Flex Premium license.

The Flex Advanced and Premium licenses for the EX Series platforms are class based, determined by the number of access ports on the switch. Class 1 (C1) switches have 12 ports, Class 2 (C2) switches have 24 Ports, and Class 3 (C3) switches have 32 or 48 Ports.

The EX2300-C switches support both subscription and perpetual Flex licenses. Subscription licenses are offered for three- and five-year terms. In addition to Junos features, the Flex Advanced and Premium subscription licenses include Juniper Wired Assurance. Flex Advanced and Premium subscription licenses also allow portability across the same tier and class of switches, ensuring investment protection for the customer.

For a complete list of features supported by the Flex Standard, Advanced, and Premium tiers, or to learn more about EX Series licenses, please visit juniper.net/documentation/us/en/software/license/licensing/topics/concept/ flex-licenses-for-ex.html.

Product options

Table 1. EX2300-C Switch models

Model	Access Port Configuration	Uplink Ports	PoE+ Capable Ports	Height	PoE+ Budget	Power Supply Rating
EX2300-C-12T	12-port 10/100/1000BASE-T	Two SFP/SFP+ ports	0	1 RU	N/A	40 W AC
EX2300-C-12P	12-port 10/100/1000BASE-T	Two SFP/SFP+ ports	12	1 RU	124 W	170 W AC

EX2300-C

Enhanced limited lifetime warranty

The EX2300-C switches include an enhanced limited lifetime hardware warranty that provides next business day advance hardware replacement for as long as the original purchaser owns the product. The warranty includes lifetime software updates, advanced shipping of spares within one business day, and 24x7 Juniper Networks Technical Assistance Center (JTAC) support for 90 days after the purchase date. Power supplies and fan trays are covered for a period of five years. For complete details, please visit support.juniper.net/support/.

Physical specifications

Power options

— Fixed internal power supply (AC)

Model	Max. System Power Consumption (Input Power without PoE)	Total PoE Power Budget
EX2300-C-12T	Power without PoE)	0
EX2300-C-12P	24 W AC	124W

Dimensions (W x H x D)

— EX2300-C-12T: 10.98 x 1.72 x 9.4 in (27.9 x 4.4 x 23.9 cm)

— EX2300-C-12P: 10.98 x 1.72 x 9.4 in (27.9 x 4.4 x 23.9 cm)

Backplane

— 40 Gbps Virtual Chassis interconnect to link up to four switches as a single logical device

System weight

— EX2300-C-12T: 5.45 lb (2.48 kg)

— EX2300-C-12P: 6.99 lb (3.17 kg)

Environmental ranges

— Operating temperature: 32° to 104° F (0° to 40° C)[1, 2]

— Storage temperature: –40° to 158° F (–40° to 70° C)

— Operating altitude: up to 5,000 ft (1524 m)

— Nonoperating altitude: up to 16,000 ft (4877 m)

— Relative humidity operating: 10% to 85% (noncondensing)

— Relative humidity nonoperating: 0% to 95% (noncondensing)

Cooling

— Fanless operation

Hardware specifications

Switching Engine Model

— Store and forward

DRAM

— 2 GB with ECC

Flash

— 2 GB

CPU

— 1.25 GHz ARM CPU

GbE port density per system

— 14 (12 access ports + 2 uplink ports)

Physical layer

— Physical port redundancy: Redundant trunk group (RTG)

— Cable diagnostics for detecting cable breaks and shorts

— Auto medium-dependent interface/medium-dependent

— interface crossover (MDI/MDIX) support

— Port speed downshift/setting maximum advertised speed on 10/100/1000BASE-T ports

— Digital optical monitoring for optical ports

Packet-switching capacities (maximum with 64-byte packets)

— 32 Gbps (unidirectional)/64 Gbps (bidirectional)

Software specifications

Layer 2/Layer 3 throughput (mpps) (maximum with 64 byte packets)

— 47 Mpps (wire speed)

Layer 2 features

— Maximum MAC addresses in hardware: 16,000

— Jumbo frames: 9216 bytes

— Number of VLANs supported: 4093 (2044 active VLAN)

— Range of possible VLAN IDs: 1-4094

— Port-based VLAN

— MAC-based VLAN

— Voice VLAN

— Layer 2 Protocol Tunneling (L2PT)

— IEEE 802.1ak: Multiple VLAN Registration Protocol (MVRP)

— Compatible with Per-VLAN Spanning Tree Plus (PVST+)

— RVI (Routed VLAN Interface)

— IEEE 802.1AB: Link Layer Discovery Protocol (LLDP)

— LLDP-MED with VoIP integration

[1] To support operating temperature ranges beyond 40° C at altitudes up to 5,000 feet, extended temperature-range SFPs should be used when fiber uplinks are being utilized.

[2] Use extended temperature-grade transceivers for optical reach of ER and ZR.

— IEEE 802.1ad Q-in-Q tunneling
— IEEE 802.1br Bridge Port Extension
— IEEE 802.1D: Spanning Tree Protocol
— IEEE 802.1p: CoS Prioritization
— IEEE 802.1Q: VLAN Tagging
— IEEE 802.1Q-in-Q: VLAN Stacking
— IEEE 802.1s: Multiple Spanning Tree Protocol (MSTP)
— Number of MST instances supported: 64
— IEEE 802.1w: Rapid Spanning Tree Protocol (RSTP)
— IEEE 802.1X: Port Access Control
— IE EE 802.3: 10BASE-T
— IEEE 802.3u: 100BASE-TX
— IEEE 802.3ab: 1000BASE-T
— IEEE 802.3z: 1000BASE-X
— IEEE 802.3af: PoE
— IEEE 802.3at: PoE+
— IEEE 802.3ad: Link Aggregation Control Protocol (LACP)
— IEEE 802.3x: Pause Frames/Flow Control

Layer 3 features: IPv4

— Maximum number of ARP entries: 1,500
— Maximum number of IPv4 unicast routes in hardware: 512 prefixes; 4,096 host routes
— Maximum number of IPv4 multicast routes in hardware: 2,048 groups; 2,048 multicast routes
— Routing Protocols: RIP v1/v2, OSPF v1/v2
— Static routing
— Routing policy
— Bidirectional Forwarding Detection (BFD) with slow timers (> 3 sec)
— IP directed broadcast

Layer 3 features: IPv6

— Maximum number of Neighbor Discovery (ND) entries: 1,500
— Maximum number of IPv6 unicast routes in hardware: 512 prefixes; 2,048 host routes
— Maximum number of IPv6 multicast routes in hardware: 1,024 groups; 1,024 multicast routes
— Neighbor discovery, system logging, Telnet, SSH, SNMP, Network Time Protocol (NTP), Domain Name System (DNS)
— Static routing
— Routing protocols: RIPng, OSPF v3

Access control lists (ACLs) (Junos OS firewall filters)

— Port-based ACL (PACL)—256 ingress; 256 egress
— VLAN-based ACL (VACL)—256 ingress; 256 egress
— Router-based ACL (RACL)—256 ingress; 512 egress
— ACL entries (ACE) in hardware per system: 2,000
— ACL counter for denied packets
— ACL counter for permitted packets>
— Ability to add/remove/change ACL entries in middle of list (ACL editing)
— L2-L4 ACL

Access security

— MAC limiting
— Allowed MAC addresses—configurable per port
— Sticky MAC (persistent MAC address learning)
— Dynamic ARP inspection (DAI)
— Proxy ARP
— Static ARP support
— DHCP snooping
— 802.1X port-based
— 802.1X multiple supplicants
— 802.1X with VLAN assignment
— 802.1X with authentication bypass access (based on host MAC0 address)
— 802.1X with VoIP VLAN support
— 802.1X dynamic access control list (ACL) based on RADIUS attributes
— 802.1X Supported EAP types: Message Digest 5 (MD5), Transport Layer Security (TLS), Tunneled Transport Layer Security (TTLS), Protected Extensible Authentication Protocol (PEAP)
— IPv6 RA Guard
— IPv6 Neighbor Discovery Inspection
— Captive Portal
— Static MAC authentication
— MAC-RADIUS
— Control plane DoS protection
— Fallback Authentication
— Trusted Network Connect (TNC) certified

High availability

— Link Aggregation

— 802.3ad (LACP) support:

- Number of LAGs supported: 128
- Maximum number of ports per LAG: 8

— Tagged ports support in LAG

— Uplink failure detection

Quality of service (QoS)

— Layer 2 QoS

— Layer 3 QoS

— Ingress policing: 2 rate 3 color

— Hardware queues per port: 8

— Scheduling methods (egress): Strict Priority (SP), shaped-deficit weighted round-robin (SDWRR)

— 802.1p: DSCP/IP precedence trust and marking

— L2-L4 classification criteria: Interface, MAC address, Ethertype, 802.1p, VLAN, IP address, DSCP/IP precedence, TCP/UDP port numbers

— Congestion avoidance capabilities: Tail drop and WRED

Multicast

— Internet Group Management Protocol (IGMP) snooping entries: 2,000

— IGMP: v1, v2, v3

— IGMP snooping

— PIM sparse mode (PIM SM), PIM source-specific multicast (PIM SSM), PIM dense mode (PIM DM)

Management and analytics platforms

— Juniper Wired Assurance for Campus

— Junos Space® Network Director for Campus

— Junos Space Management Applications

Device management and operations

— Junos OS CLI

— Junos Web interface (J-Web)

— Out-of-band management: Serial, 10/100BASE-T Ethernet

— ASCII configuration

— Rescue configuration

— Configuration rollback

— Image rollback

— Simple Network Management Protocol (SNMP): v1, v2c, v3

— Remote monitoring (RMON) (RFC 2819) Groups 1, 2, 3, 9

— Network Time Protocol (NTP)

— DHCP server

— DHCP client and DHCP proxy

— DHCP relay and helper

— RADIUS authentication

— TACACS+ authentication

— SSHv2

— Secure copy

— HTTP/HTTPs

— DNS resolver

— System log logging

— Temperature sensor

— Configuration backup via FTP/secure copy

— Interface range

Supported MIBs

— RFC 1155 Structure of Management Information (SMI)

— RFC 1157 SNMPv1

— RFC 1212, RFC 1213, RFC 1215 MIB-II, Ethernet-like MIB, and TRAPs

— RFC 1493 Bridge MIB

— RFC 1643 Ethernet MIB

— RFC 1724 RIPv2 MIB

— RFC 1905 RFC 1907 SNMP v2c, SMIv2, and Revised MIB-II

— RFC 1981 Path MTU Discovery for IPv6

— RFC 2011 SNMPv2 Management Information Base for the IP using SMIv2

— RFC 2012 SNMPv2 Management Information Base for the transmission Control Protocol using SMIv2

— RFC 2013 SNMPv2 Management Information Base for the User Datagram Protocol using SMIv2

— RFC 2096 IPv4 Forwarding Table MIB

— RFC 2287 System Application Packages MIB

— RFC 2460 IPv6 Specification

— RFC 2464 Transmission of IPv6 Packets over Ethernet Networks
— RFC 2570-2575 SNMPv3, User-based Security, Encryption, and Authentication
— RFC 2576 Coexistence between Version 1, Version 2, and Version 3 of the Internet-standard Network Management Framework
— RFC 2578 SNMP Structure of Management Information MIB
— RFC 2579 SNMP Textual Conventions for SMIv2
— RFC 2665 Definitions of Managed Objects for the Ethernet like Interface Types
— RFC 2819 RMON MIB
— RFC 2863 The Interfaces Group MIB
— RFC 2922 LLDP MIB
— RFC 2925 Definitions of Managed Objects for Remote Ping,
— Traceroute, and Lookup Operations
— RFC 3413 SNMP Application MIB
— RFC 3414 User-based Security Model for SNMPv3
— RFC 3415 View-based Access Control Model (VACM) for SNMP
— RFC 3484 Default Address Selection for IPv6
— RFC 3621 PoE-MIB (PoE switches only)
— RFC 4188 STP and Extensions MIB
— RFC 4213 Basic Transition Mechanisms for IPv6 Hosts and Routers
— RFC 4291 IPv6 Addressing Architecture
— RFC 4363 Definitions of Managed Objects for Bridges with Traffic Classes, Multicast Filtering, and VLAN Extensions
— RFC 4443 ICMPv6 for the IPv6 Specification
— RFC 4861 Neighbor Discovery for IPv6
— RFC 4862 IPv6 Stateless Address Autoconfiguration
— Draft - blumenthal - aes - usm - 08
— Draft - reeder - snmpv3 - usm - 3desede -0

Troubleshooting

— Debugging: CLI via console, telnet, or SSH
— Diagnostics: Show and debug command statistics
— Traffic mirroring (port)
— Traffic mirroring (VLAN)
— Filter-based mirroring
— Mirroring destination ports per system: 4
— LAG port monitoring
— Multiple destination ports monitored to 1 mirror (N:1)
— Maximum number of mirroring sessions: 4
— Mirroring to remote destination (over L2): 1 destination VLAN
— Encapsulated Remote Switched Port Analyzer (ERSPAN)
— IP tools: Extended ping and trace
— HPE Juniper Networking commit and rollback

Safety certifications

— UL-UL60950-1 (Second Edition)
— C-UL to CAN/CSA 22.2 No.60950-1 (Second Edition)
— TUV/GS to EN 60950-1 (Second Edition)
— CB-IEC60950-1 (Second Edition with all country deviations)
— EN 60825-1 (Second Edition)

Electromagnetic compatibility certifications

— FCC 47CFR Part 15 Class A
— EN 55022 Class A
— ICES-003 Class A
— VCCI Class A
— AS/NZS CISPR 22 Class A
— CISPR 22 Class A
— EN 55024
— EN 300386
— CE

Telecom quality management

— TL9000

Telco

— CLEI code

Noise specifications

— 0 db (fanless)

Warranty

— Enhanced limited lifetime switch hardware warranty

Ordering information

Product number	Description
Switches	
EX2300-C-12T	EX2300 Compact Fanless 12-port 10/100/1000BASE-T, 2 X 1/10GbE SFP/SFP+ (optics sold separately)
EX2300-C-12T-VC	EX2300 Compact Fanless 12-port 10/100/1000BASE-T, 2 X 1/10GbE SFP/SFP+ with Virtual Chassis License (optics sold separately)
EX2300-C-12P	EX2300 Compact Fanless 12-port 10/100/1000BASE-T PoE+, 2 X 1/10GbE SFP/SFP+ (optics sold separately)
EX2300-C-12P-VC	EX2300 Compact Fanless 12-port 10/100/1000BASE-T PoE+, 2 X 1/10GbE SFP/SFP+ with Virtual Chassis License (optics sold separately)
EX2300-C-12T-TAA	EX2300 TAA Compact Fanless 12-port 10/100/1000BASE-T, 2 X 1/10GbE SFP/SFP+ (optics sold separately)
EX2300-C-12P-TAA	EX2300 TAA Compact Fanless 12-port 10/100/1000BASE-T PoE +, 2 X 1/10GbE SFP/SFP+ (optics sold separately)
Accessories	
EX-CBL-CON-USB	Console cable with USB Type A and mini-B connectors
EX2300-C-CBL-GRD	Cable guard
EX2300-C-MGNT-MNT	Magnet mount
EX2300-C-RMK	Rack-mount kit
Subscription licenses	
S-EX-A-C1-3	Software, EX Series Advanced license, Class 1 (12 ports), includes Wired Assurance subscription for EX Series 12-port switches, 3 year
S-EX-A-C1-5	Software, EX Series Advanced license, Class 1 (12 ports), includes Wired Assurance subscription for EX Series 12-port switches, 5 year
S-EX-A-C1-3-COR	Software, EX Series Advanced license, Class 1 (12 ports), includes Wired Assurance subscription for EX Series 12-port switches with SVC CORE support, 3 year
S-EX-A-C1-5-COR	Software, EX Series Advanced license, Class 1(12 ports), includes Wired Assurance subscription for EX Series 12-port switches with SVC CORE support, 5 year

Perpetual licenses	
S-EX-A-C2-P	Software, EX Series Advanced license, Class 2 Perpetual license for 12-port switches
EX-12-EFL	Enhanced Feature License for EX2300 12 port switches
Optics	
EX-SFP-10GE-USR	SFP+ 10-Gigabit Ethernet Ultra Short Reach Optics, 850 nm for 10m on OM1, 20m on OM2, 100m on OM3 multimode fiber
EX-SFP-10GE-DAC-1M	SFP+ 10-Gigabit Ethernet Direct Attach Copper (twinax copper cable), 1m
EX-SFP-10GE-SR	SFP+ 10GBASE-LR; LC connector; 1310 nm; 10km reach on single-mode fiber
EX-SFP-10GE-LR	SFP+ 10GBASE-LR; LC connector; 1310 nm; 10km reach on single-mode fiber
EX-SFP-10GE-DAC-3M	SFP+ 10-Gigabit Ethernet Direct Attach Copper (twinax copper cable), 3m
EX-SFP-10GE-DAC-5M	SFP+ 10-Gigabit Ethernet Direct Attach Copper (twinax copper cable), 5m
EX-SFP-1GE-SX	SFP 1000BASE-SX; LC connector; 850 nm; 550m reach on multimode fiber
EX-SFP-1GE-SX-ET	Extended Temperature SFP 1000BASE-SX; LC connector; 850
EX-SFP-1GE-SX-ET	Extended Temperature SFP 1000BASE-SX; LC connector; 850
EX-SFP-1GE-LX	SFP 1000BASE-LX; LC connector; 1310 nm; 10 km reach on single-mode fiber
EX-SFP-1GE-LH	SFP 1000BASE-LH Gigabit Ethernet Optics, 1550 nm for 70 km transmission on single-mode fiber
EX-SFP-1GE-T	SFP 10/100/1000BASE-T Copper Transceiver Module for up to 100m transmission on Category[1]
EX-SFP-1GE-LX40K	SFP 1000BASE-LX Gigabit Ethernet Optics, 1310 nm for 40 km transmission on single-mode fiber
EX-SFP-GE10KT13R14	SFP 1000BASE-BX Gigabit Ethernet Optics, Tx 1310 nm/Rx 1490 nm for 10 km transmission on single strand of single-mode fiber
EX-SFP-GE10KT14R13	SFP 1000BASE-BX Gigabit Ethernet Optics, Tx 1490 nm/Rx 1310 nm for 10 km transmission on single strand of single-mode fiber
EX-SFP-GE10KT13R15	SFP 1000BASE-BX Gigabit Ethernet Optics, Tx 1310 nm/Rx 1550 nm for 10 km transmission on single strand of single-mode fiber
EX-SFP-GE10KT15R13	SFP 1000BASE-BX Gigabit Ethernet Optics, Tx 1550 nm/Rx 1310 nm for 10 km transmission on single strand of single-mode fiber

[1] Each switch comes with a RJ-45-to-DB-9 serial port adapter, power cord retainer, and four rubber feet. Each system also ships with a power cord for the country to which it is shipped

Ordering information (continued)

Product number	Description
EX-SFP-GE40KT13R15	SFP 1000BASE-BX Gigabit Ethernet Optics, Tx 1310 nm/Rx 1550 nm for 40 km transmission on single strand of single-mode fiber
EX-SFP-GE40KT15R13	SFP 1000BASE-BX Gigabit Ethernet Optics, Tx 1550 nm/Rx 1310 nm for 40 km transmission on single strand of single-mode fiber
EX-SFP-GE80KCW1470	SFP Gigabit Ethernet CWDM, LC connector; 1470 nm, 80 km reach on single-mode fiber
EX-SFP-GE80KCW1490	SFP Gigabit Ethernet CWDM, LC connector; 1490 nm, 80 km reach on single-mode fiber
EX-SFP-GE80KCW1510	SFP Gigabit Ethernet CWDM, LC connector; 1510 nm, 80 km reach on single-mode fiber
EX-SFP-GE80KCW1530	SFP Gigabit Ethernet CWDM, LC connector; 1530 nm, 80 km reach on single-mode fiber
EX-SFP-GE80KCW1550	SFP Gigabit Ethernet CWDM, LC connector; 1550 nm, 80 km reach on single-mode fiber
EX-SFP-GE80KCW1570	SFP Gigabit Ethernet CWDM, LC connector; 1570 nm, 80 km reach on single-mode fiber
EX-SFP-GE80KCW1590	SFP Gigabit Ethernet CWDM, LC connector; 1590 nm, 80 km reach on single-mode fiber
EX-SFP-GE80KCW1610	SFP Gigabit Ethernet CWDM, LC connector; 1610 nm, 80 km reach on single-mode fiber

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